> $eq1 := diff(x(t), t) + x(t) = \frac{2}{\mathrm{sqrt}(\mathrm{Pi})} \cdot \exp(-t^2 - t)$

$$eq1 := \frac{\mathrm{d}}{\mathrm{d}t} x(t) + x(t) = \frac{2\, \mathrm{e}^{-t2-t}}{\sqrt{\pi}} \quad (1)$$

> $dsolve(eq1, x(t))$

$$x(t) = (\mathrm{erf}(t) + c_1)\, \mathrm{e}^{-t} \quad (2)$$

> $int(\exp(t^2), t)$

$$\frac{\sqrt{\pi}\ \mathrm{erfi}(t)}{2} \quad (3)$$

> $int\left(\frac{2}{\mathrm{sqrt}(\mathrm{Pi})} \cdot \exp(-t^2), t\right)$

$$\mathrm{erf}(t) \quad (4)$$

> $eq2 := diff(x(t), t\$2) + 3 \cdot diff(x(t), t) + x(t) = 1$

$$eq2 := \frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) + 3 \frac{\mathrm{d}}{\mathrm{d}t} x(t) + x(t) = 1 \quad (5)$$

> $sol2 := dsolve(eq2, x(t))$

$$sol2 := x(t) = \mathrm{e}^{\frac{(\sqrt{5}-3)\, t}{2}} c_2 + \mathrm{e}^{-\frac{(3+\sqrt{5})\, t}{2}} c_1 + 1 \quad (6)$$

> $limit(rhs(sol2), t = \mathrm{infinity})$

$$1 \quad (7)$$

> $eq3 := diff(x(t), t\$2) + 4 \cdot x(t) = 1$

$$eq3 := \frac{\mathrm{d}^2}{\mathrm{d}t^2} x(t) + 4\, x(t) = 1 \quad (8)$$

> $ic3 := x(0) = \frac{5}{4}, \mathrm{D}(x)(0) = 0$

$$ic3 := x(0) = \frac{5}{4}, \mathrm{D}(x)(0) = 0 \quad (9)$$

> $sol3 := dsolve(\{eq3, ic3\}, x(t))$

$$sol3 := x(t) = \frac{1}{4} + \cos(2\, t) \quad (10)$$

> $eval(rhs(sol3), t = \mathrm{Pi})$

$$\frac{5}{4} \quad (11)$$

> $eq4 := diff(x(t), t) = 3 \cdot x(t) + t^3$

$$eq4 := \frac{\mathrm{d}}{\mathrm{d}t} x(t) = 3\, x(t) + t^3 \quad (12)$$

> $sol4 := dsolve(eq4, x(t))$

(13)

$$sol4 := x(t) = -\frac{t^2}{3} - \frac{t^3}{3} - \frac{2\,t}{9} - \frac{2}{27} + e^{3\,t}\,c_1 \tag{13}$$

```
> subs(_C1 = 0, rhs(sol4))
```

$$-\frac{1}{3}\,t^2 - \frac{1}{3}\,t^3 - \frac{2}{9}\,t - \frac{2}{27} \tag{14}$$

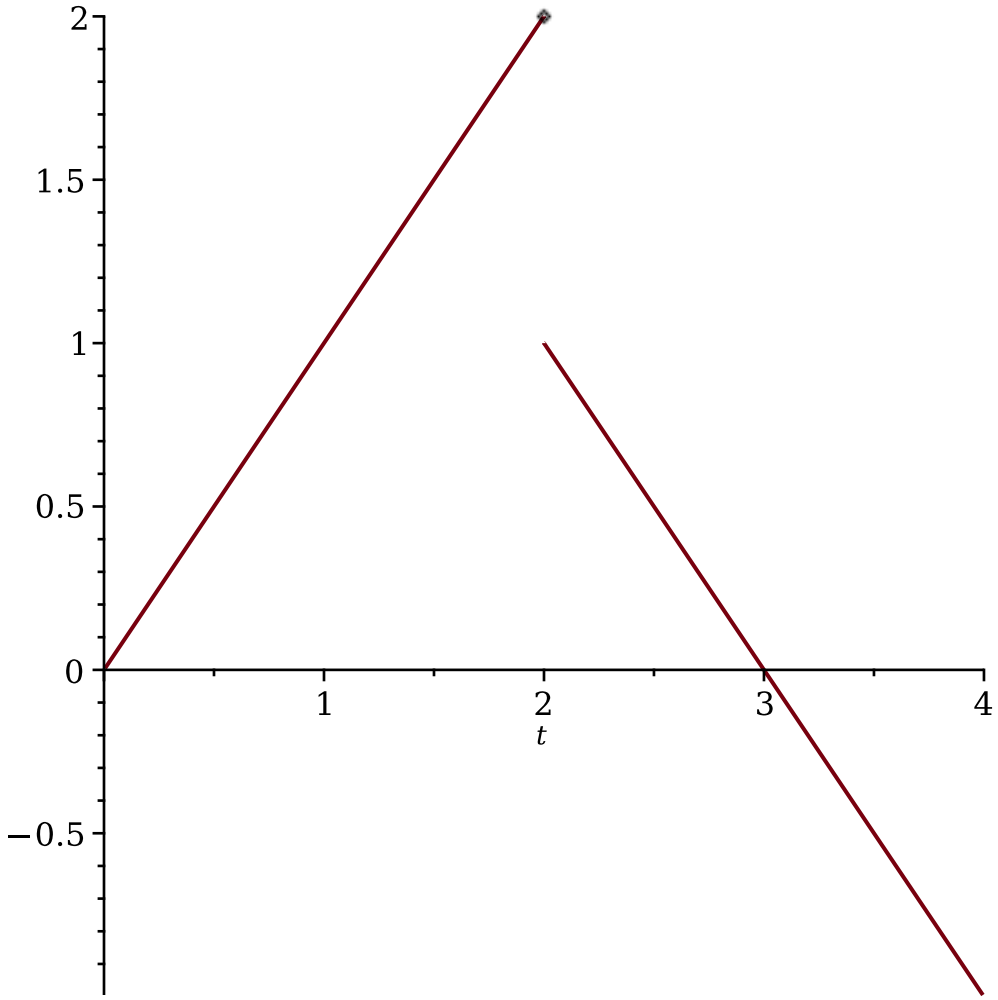
```
> f5 := piecewise(t <= 2, t, t > 2, 3 - t)
```

$$f5 := \begin{cases} t & t \le 2 \\ 3 - t & 2 < t \end{cases} \tag{15}$$

```
> with(plots)
```

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*] (16)

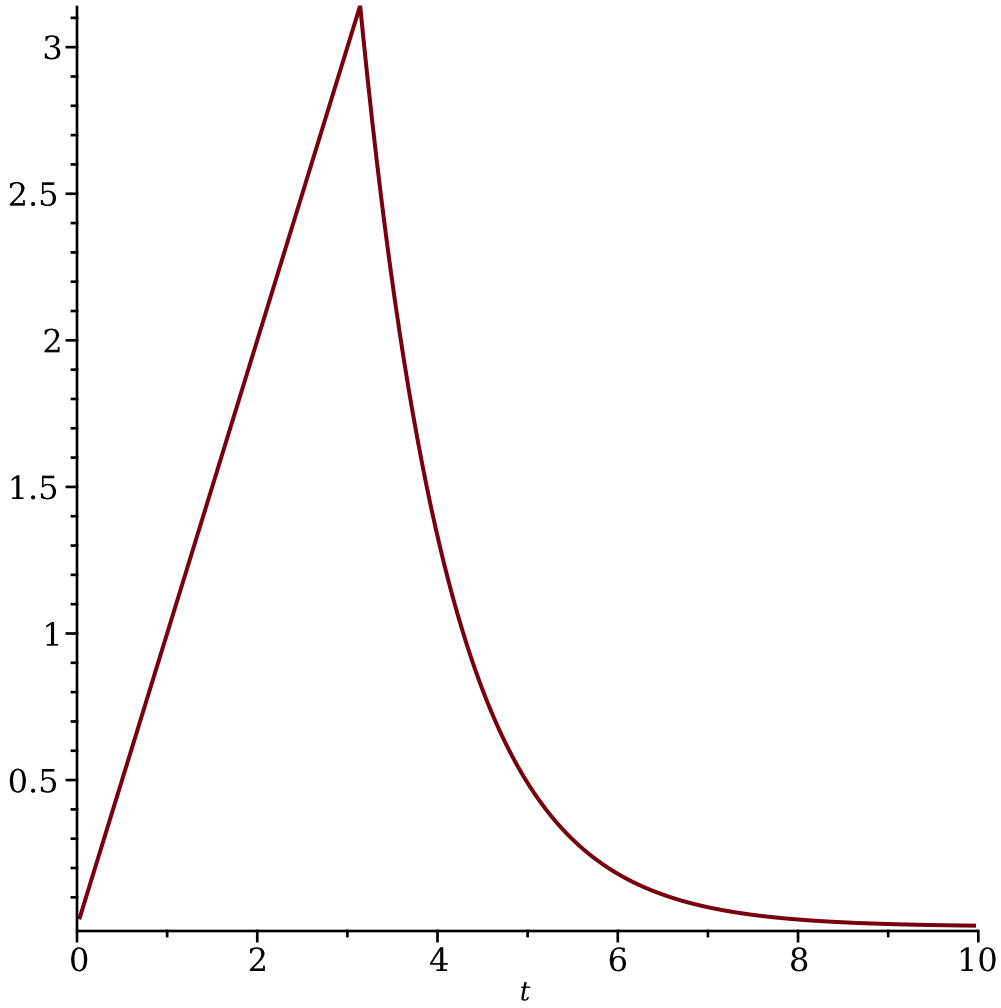
```
> plot(f5, t = 0 .. 4, discont = true)
```




```
> f6 := piecewise(t <= Pi, t, t > Pi, Pi*exp(Pi - t))
```

$$f6 := \begin{cases} t & t \le \pi \\ \pi\, e^{\pi - t} & \pi < t \end{cases} \quad (17)$$

```
> plot(f6, t = 0 .. 10, discont = true)
```



> $eq7 := diff(x(t), t\$2) + x(t) = f6$

$$eq7 := \frac{d^2}{dt^2} x(t) + x(t) = \begin{cases} t & t \le \pi \\ \pi e^{\pi - t} & \pi < t \end{cases} \quad (18)$$

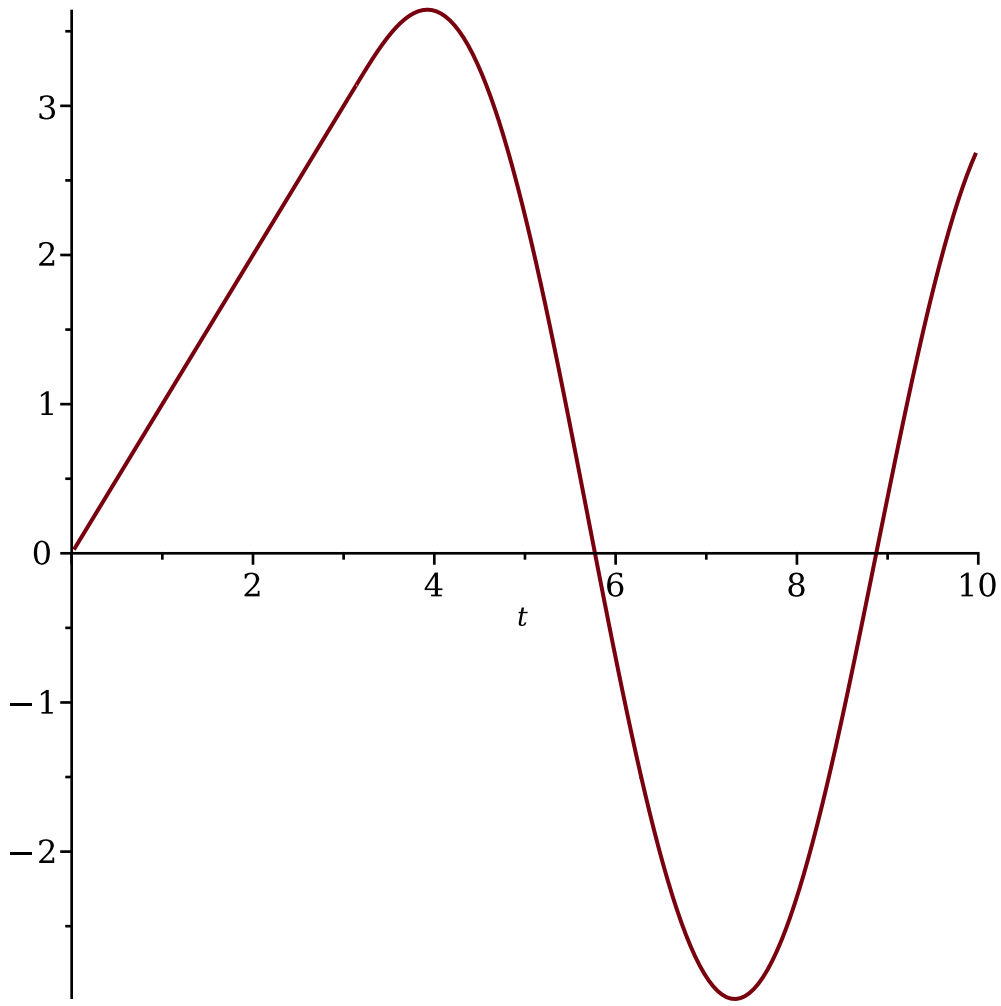
> $ic7 := x(0) = 0, D(x)(0) = 1$

$$ic7 := x(0) = 0, D(x)(0) = 1 \quad (19)$$

> $sol7 := dsolve(\{eq7, ic7\}, x(t))$

$$sol7 := x(t) = \begin{cases} t & t < \pi \\ -\frac{\sin(t)\,\pi}{2} - \frac{\cos(t)\,\pi}{2} + \frac{\pi e^{\pi - t}}{2} - \sin(t) & \pi \le t \end{cases} \quad (20)$$

> $plot(rhs(sol7), t = 0..10)$



> $eq8 := diff(x(t), t\$2) + x(t) = \cos(\text{omega}\cdot t)$

$$eq8 := \frac{d^2}{dt^2} x(t) + x(t) = \cos(\omega t) \quad (21)$$

> $ic8 := x(0) = 0, D(x)(0) = 0$

$$ic8 := x(0) = 0, D(x)(0) = 0 \quad (22)$$

> $sol8 := dsolve(\{eq8, ic8\}, x(t))$

$$sol8 := x(t) = \frac{\cos(t) - \cos(\omega t)}{\omega^2 - 1} \quad (23)$$

> $eq81 := diff(x(t), t\$2) + x(t) = \cos(t)$

$$eq81 := \frac{d^2}{dt^2} x(t) + x(t) = \cos(t) \quad (24)$$

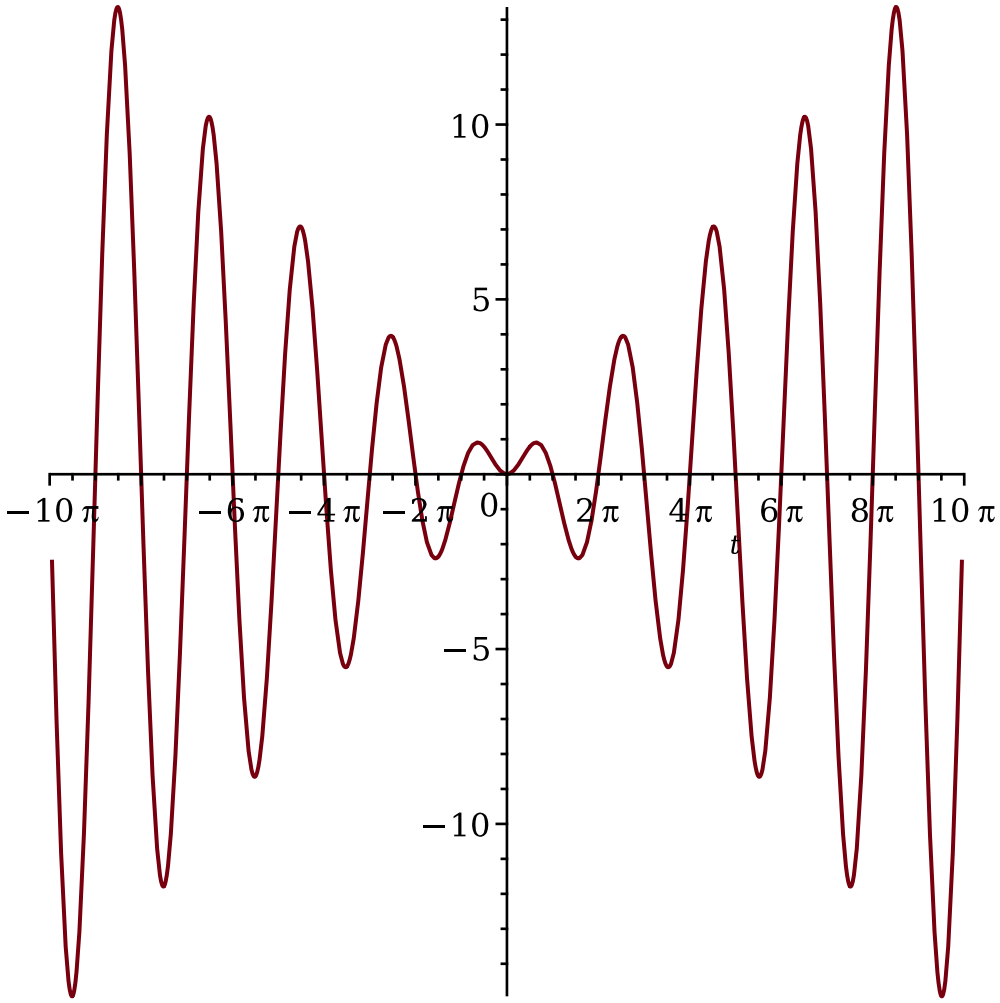
> $sol81 := dsolve(\{eq81, ic8\}, x(t))$

$$sol81 := x(t) = \frac{\sin(t)\, t}{2} \quad (25)$$

> $limit(rhs(sol8), \text{omega} = 1)$

$$\frac{\sin(t)\, t}{2} \quad (26)$$

> $plot(rhs(sol81), t = -10\ \mathrm{Pi}..10\ \mathrm{Pi})$



> $eq9 := diff(x(t), t\$2) - 4 \cdot x(t) = \exp(\mathrm{alpha} \cdot t)$

$$eq9 := \frac{d^2}{dt^2} x(t) - 4\,x(t) = e^{\alpha t} \tag{27}$$

> $ic9 := x(0) = 0, D(x)(0) = 0$

$$ic9 := x(0) = 0, D(x)(0) = 0 \tag{28}$$

> $sol9 := dsolve(\{eq9, ic9\}, x(t))$

$$sol9 := x(t) = \frac{(\alpha - 2)\, e^{-2t} + (-\alpha - 2)\, e^{2t} + 4\, e^{\alpha t}}{4\,\alpha^2 - 16} \tag{29}$$

> $eq92 := diff(x(t), t\$2) - 4 \cdot x(t) = \exp(2 \cdot t)$

$$eq92 := \frac{d^2}{dt^2} x(t) - 4\,x(t) = e^{2t} \tag{30}$$

> $sol92 := dsolve(\{eq92, ic9\}, x(t))$

$$sol92 := x(t) = \frac{(4t - 1)\, e^{2t}}{16} + \frac{e^{-2t}}{16} \tag{31}$$

> $limit(rhs(sol9), \mathrm{alpha} = 2)$

(32)

$$\frac{e^{-2t}}{16} - \frac{e^{2t}}{16} + \frac{t e^{2t}}{4} \tag{32}$$

> $f10 := piecewise\left(0 \le t < \frac{Pi}{2}, t, \frac{Pi}{2} \le t \le Pi, Pi - t, t > Pi, 0\right)$

$$f10 := \begin{cases} t & 0 \le t < \frac{\pi}{2} \\ \pi - t & \frac{\pi}{2} \le t \le \pi \\ 0 & \pi < t \end{cases} \tag{33}$$

> $eq10 := diff(x(t), t\$2) + x(t) = f10$

$$eq10 := \frac{d^2}{dt^2} x(t) + x(t) = \begin{cases} t & 0 \le t < \frac{\pi}{2} \\ \pi - t & \frac{\pi}{2} \le t \le \pi \\ 0 & \pi < t \end{cases} \tag{34}$$

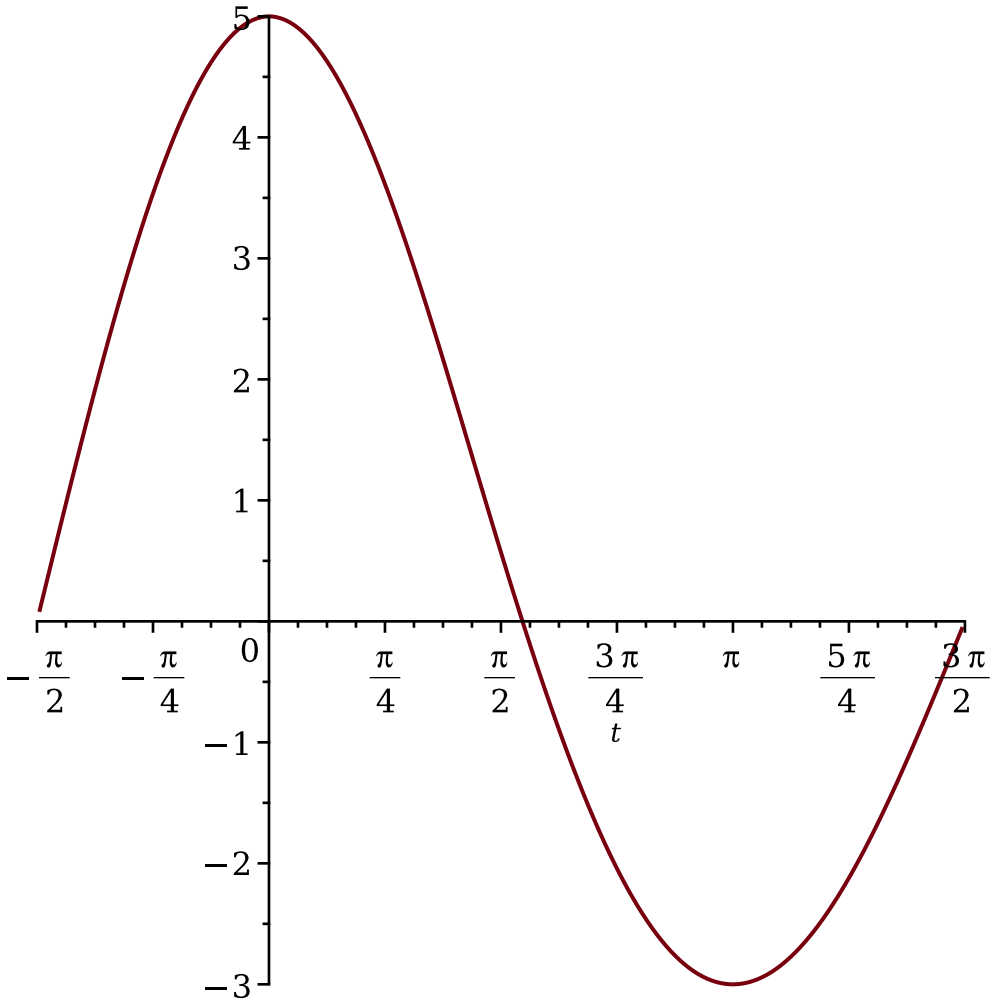
> $ic10 := x(0) = 5, D(x)(0) = 0$

$$ic10 := x(0) = 5, D(x)(0) = 0 \tag{35}$$

> $sol10 := dsolve(\{eq10, ic10\}, x(t))$

$$sol10 := x(t) = \begin{cases} 5\cos(t) & t < 0 \\ 5\cos(t) + t - \sin(t) & t < \frac{\pi}{2} \\ 3\cos(t) + \pi - t - \sin(t) & t < \pi \\ 3\cos(t) & \pi \le t \end{cases} \tag{36}$$

> $plot(rhs(sol10))$



> $sys11 := \{diff(x(t), t) = -2 \cdot x(t), diff(y(t), t) = -3 \cdot y(t), x(0) = 1, y(0) = 1\}$

$$sys11 := \left\{ \frac{d}{dt} x(t) = -2\,x(t),\ \frac{d}{dt} y(t) = -3\,y(t),\ x(0) = 1,\ y(0) = 1 \right\} \quad (37)$$

> $sol11 := dsolve(sys11, \{x(t), y(t)\})$

$$sol11 := \{x(t) = e^{-2t},\ y(t) = e^{-3t}\} \quad (38)$$

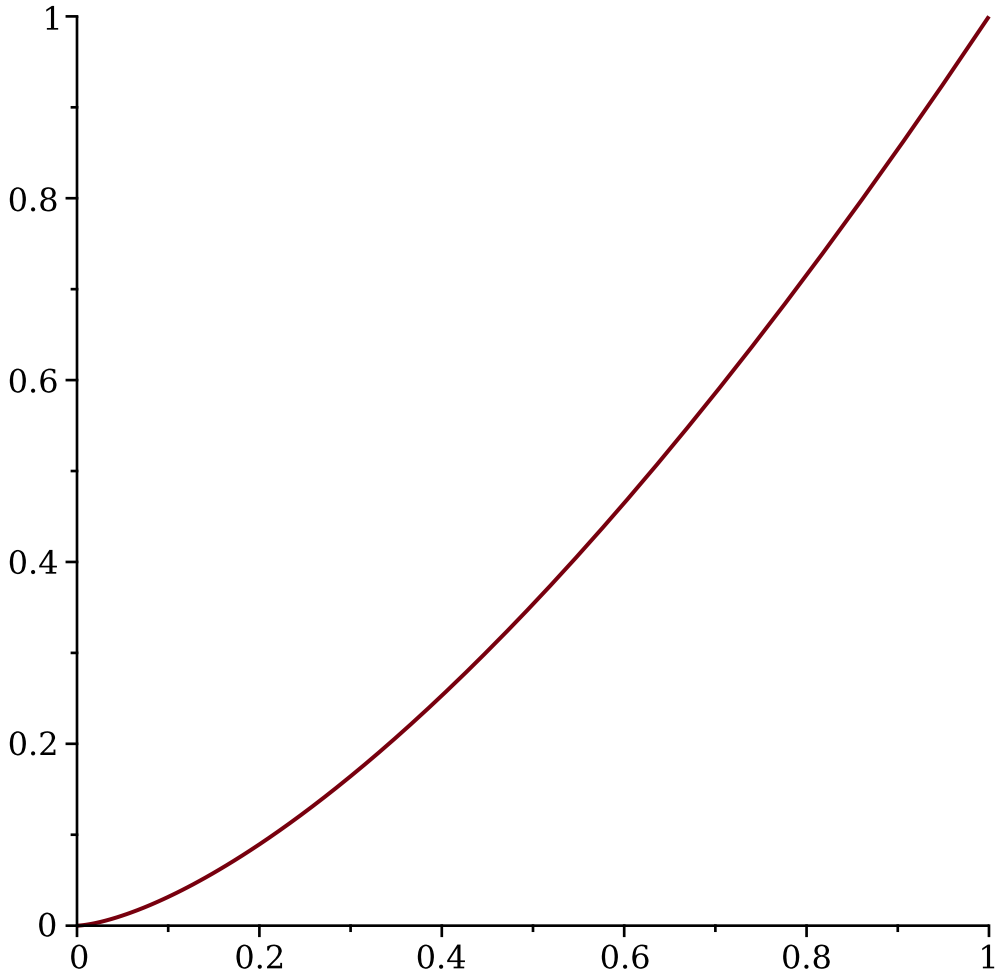
> $xsol11 := rhs(sol11[1])$

$$xsol11 := e^{-2t} \quad (39)$$

> $ysol11 := rhs(sol11[2])$

$$ysol11 := e^{-3t} \quad (40)$$

> $plot([xsol11, ysol11, t = 0..5])$



> $sys12 := \{diff(x(t), t) = -2 \cdot x(t), diff(y(t), t) = 3 \cdot y(t), x(0) = 1, y(0) = 1\}$

$$sys12 := \left\{ \frac{d}{dt} x(t) = -2\,x(t),\ \frac{d}{dt} y(t) = 3\,y(t),\ x(0) = 1,\ y(0) = 1 \right\} \quad (41)$$

> $sol12 := dsolve(sys12, \{x(t), y(t)\})$

$$sol12 := \{x(t) = e^{-2\,t},\ y(t) = e^{3\,t}\} \quad (42)$$

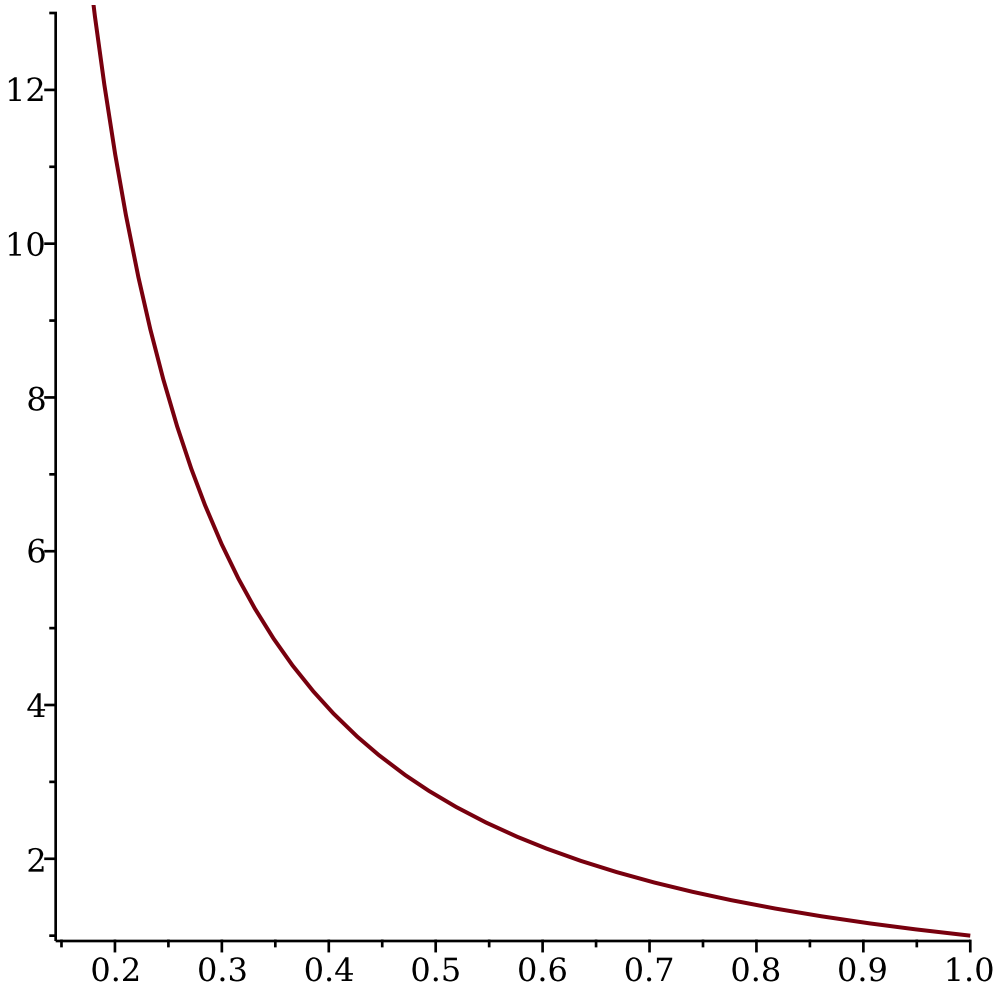
> $xsol12 := rhs(sol12[1])$

$$xsol12 := e^{-2\,t} \quad (43)$$

> $ysol12 := rhs(sol12[2])$

$$ysol12 := e^{3\,t} \quad (44)$$

> $plot([xsol12, ysol12, t = 0..5])$



> $sys13 := \{diff(x(t), t) = -y(t), diff(y(t), t) = 4 \cdot x(t), x(0) = 1, y(0) = 1\}$

$$sys13 := \left\{ \frac{d}{dt} x(t) = -y(t), \frac{d}{dt} y(t) = 4\,x(t), x(0) = 1, y(0) = 1 \right\} \quad (45)$$

> $sol13 := dsolve(sys13, \{x(t), y(t)\})$

$$sol13 := \left\{ x(t) = -\frac{\sin(2\,t)}{2} + \cos(2\,t), y(t) = \cos(2\,t) + 2\sin(2\,t) \right\} \quad (46)$$

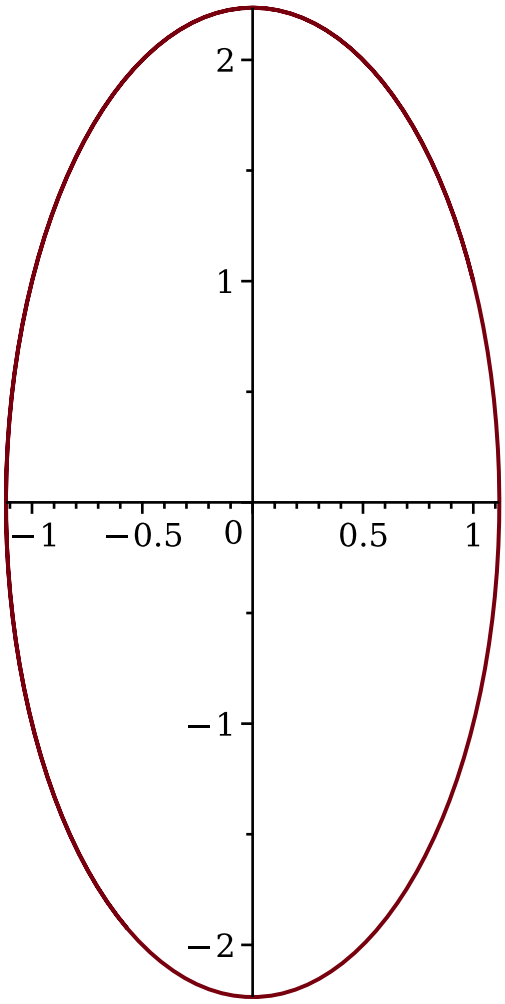
> $xsol13 := rhs(sol13[1])$

$$xsol13 := -\frac{\sin(2\,t)}{2} + \cos(2\,t) \quad (47)$$

> $ysol13 := rhs(sol13[2])$

$$ysol13 := \cos(2\,t) + 2\sin(2\,t) \quad (48)$$

> $plot([xsol13, ysol13, t = 0..5], scaling = constrained)$



> $sys14 := \{diff(x(t), t) = -x(t) + 3 \cdot y(t), diff(y(t), t) = -3 \cdot x(t) - y(t), x(0) = 1, y(0) = 1\}$

$$sys14 := \left\{ \frac{d}{dt} x(t) = -x(t) + 3\,y(t),\ \frac{d}{dt} y(t) = -3\,x(t) - y(t),\ x(0) = 1,\ y(0) = 1 \right\} \quad (49)$$

> $sol14 := dsolve(sys14, \{x(t), y(t)\})$

$$sol14 := \{x(t) = e^{-t}(\cos(3\,t) + \sin(3\,t)),\ y(t) = e^{-t}(\cos(3\,t) - \sin(3\,t))\} \quad (50)$$

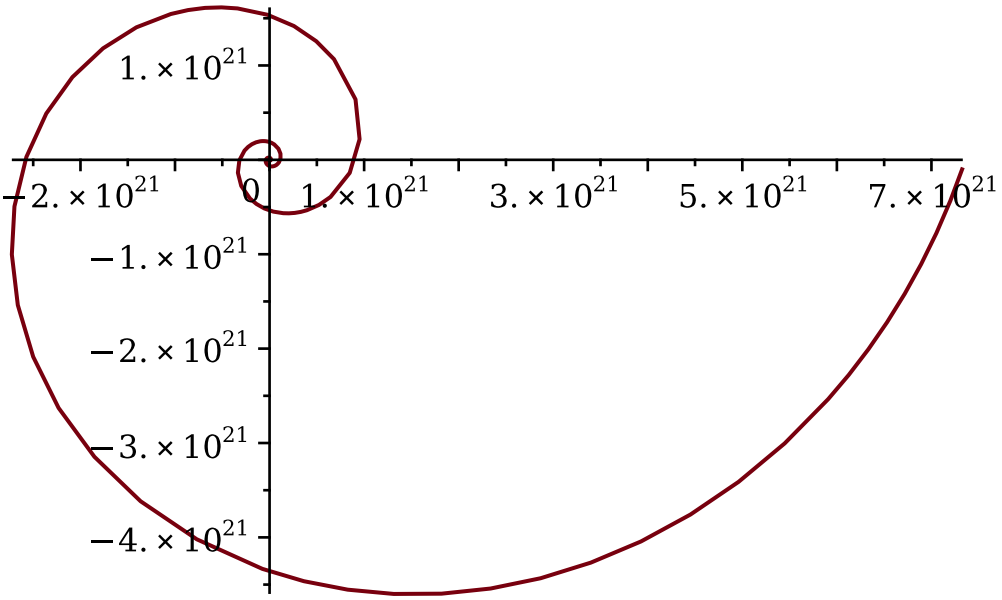
> $xsol14 := rhs(sol14[1])$

$$xsol14 := e^{-t}(\cos(3\,t) + \sin(3\,t)) \quad (51)$$

> $ysol14 := rhs(sol14[2])$

$$ysol14 := e^{-t}(\cos(3\,t) - \sin(3\,t)) \quad (52)$$

> $plot([xsol14, ysol14, t = -50..50], scaling = constrained)$



> $sys15 := \{diff(x(t), t) = -x(t) + 3\cdot y(t), diff(y(t), t) = 3\cdot x(t) - y(t), x(0) = 1, y(0) = 1\}$

$$sys15 := \left\{\frac{d}{dt}x(t) = -x(t) + 3\,y(t), \frac{d}{dt}y(t) = 3\,x(t) - y(t), x(0) = 1, y(0) = 1\right\} \quad (53)$$

> $sol15 := dsolve(sys15, \{x(t), y(t)\})$

$$sol15 := \{x(t) = e^{2\,t}, y(t) = e^{2\,t}\} \quad (54)$$

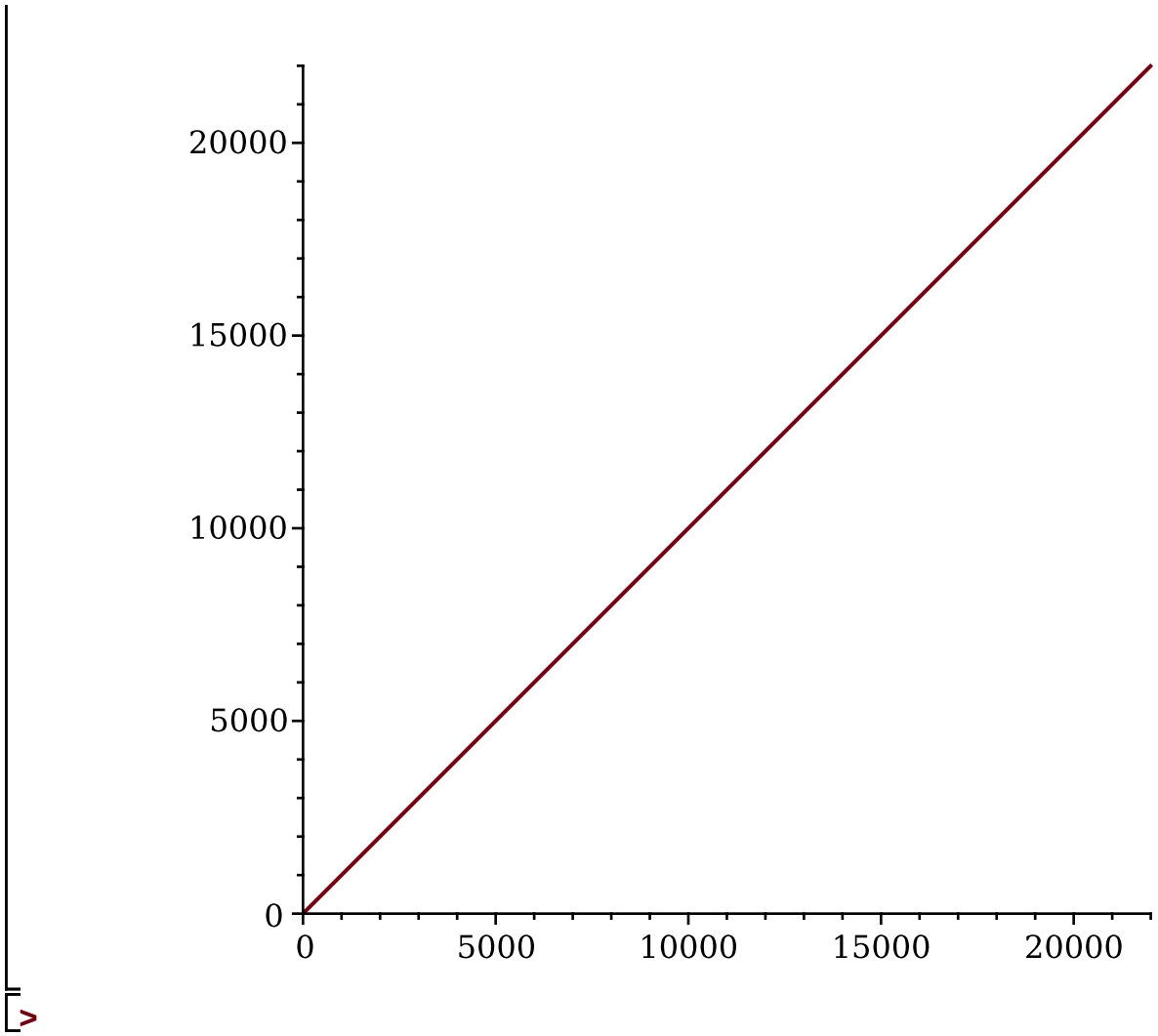
> $xsol15 := rhs(sol15[1])$

$$xsol15 := e^{2\,t} \quad (55)$$

> $ysol15 := rhs(sol15[2])$

$$ysol15 := e^{2\,t} \quad (56)$$

> $plot([xsol15, ysol15, t = -5..5], scaling = constrained)$


20000
15000
10000
5000
0
0
5000
10000
15000
20000